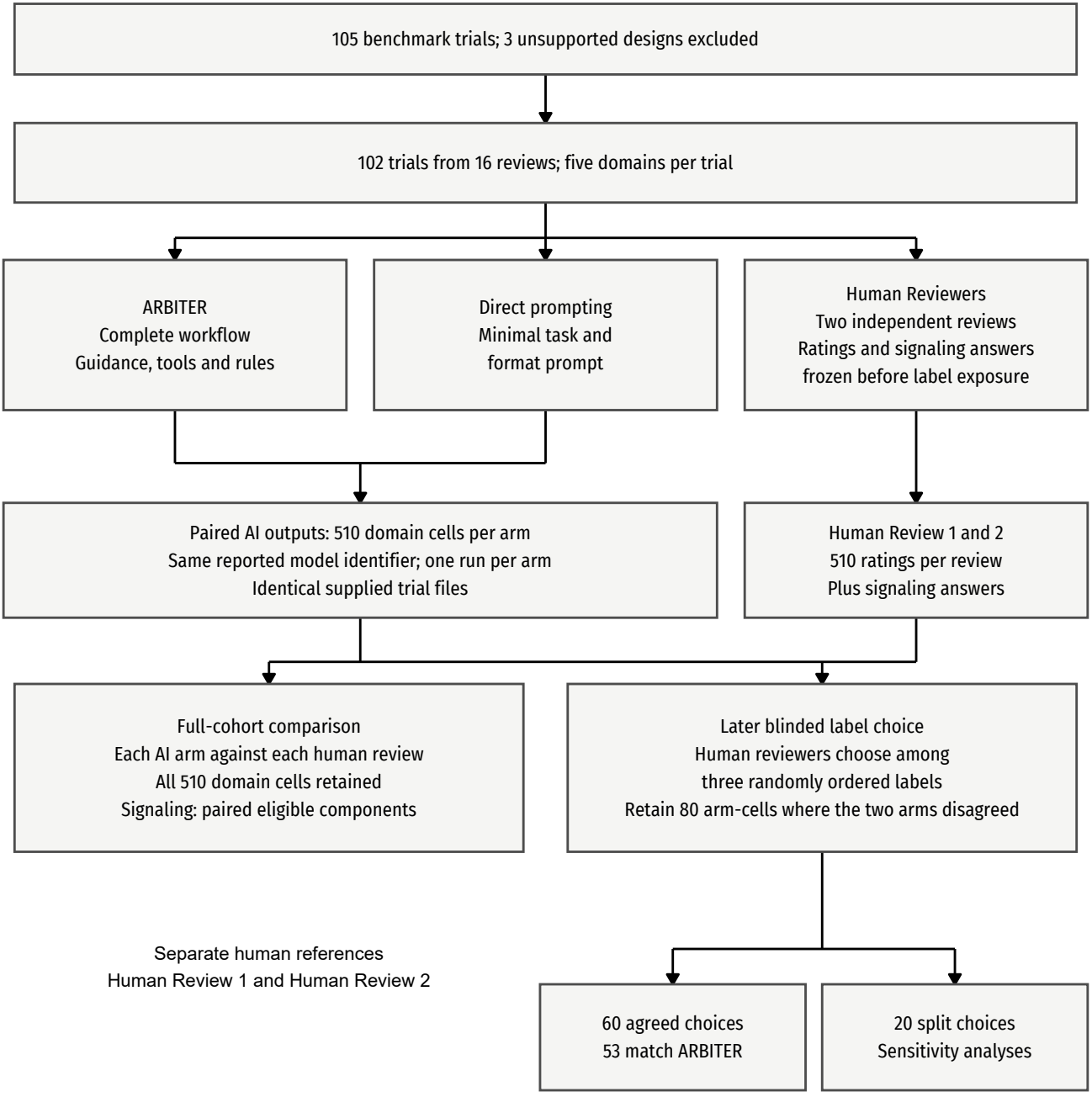
105 benchmark trials; 3 unsupported designs excluded
102 trials from 16 reviews; five domains per trial
ARBITER
Complete workflow
Guidance, tools and rules
Direct prompting
Minimal task and
format prompt
Human Reviewers
Two independent reviews
Ratings and signaling answers
frozen before label exposure
Paired AI outputs: 510 domain cells per arm
Same reported model identifier; one run per arm
Identical supplied trial files
Human Review 1 and 2
510 ratings per review
Plus signaling answers
Full-cohort comparison
Each AI arm against each human review
All 510 domain cells retained
Signaling: paired eligible components
Later blinded label choice
Human reviewers choose among
three randomly ordered labels
Retain 80 arm-cells where the two arms disagreed
Separate human references
Human Review 1 and Human Review 2
60 agreed choices
53 match ARBITER
20 split choices
Sensitivity analyses